to what is presented in this book: we have chosen a topic that should be familiar to readers with some background on algorithms, providing a review of heaps together with some new insight on branching factors.

To this end, however, we assume that the reader is familiar with some basic concepts traditionally taught in CS 101 courses: big-O notation, the RAM model, and simple data structures like arrays, lists, trees. These building blocks will be leveraged throughout the book to build increasingly complex structures and algorithms, and it's important that you familiarize yourself with such concepts, in order to be able to go through the next chapters; hence we provided a recap of these fundamental topics in the appendices at the end of the book: feel free to take a look or skim through them in order to make sure you have a good understanding of the material.

Now that we have settled the basics, we will start with the core focus of this book, and in section 2.1 we describe the structure that we will use for each of the remaining chapters.

Section 2.2 introduces the problem we are going to use in this book (how to efficiently handle events with priority), while section 2.3 outlines possible solutions, including priority queues, explaining why the latter are better than more basics data structures.

Next, in section 2.4 we describe the priority queue API[1], and we show with an example how to use it, as a black box, before delving into its internals in sections 2.5 and 2.6. In the former we analyze in detail how a d-ary heap works,